▫ 17 Statistical Inference – The t test

Consider the following examples

Example 1

A scientific experiment is conducted to determine the fat content of frozen burgers sold by two food producer and B, and in particular, to decide if there is a difference in mean fat content between the two producers.

Example 2

A survey is commissioned of first year students at WIT to determine if there is a difference between male and female students in terms of what average weekly rent they pay.

Example 3

Is the mean recovery time for patients treated with a new drug shorter than those treated with an existing drug?

Whilst the contexts of the example are different, they are fundamentally asking the same kind of question. I.e. does the mean value of a scale variable (rent amount, fat mass or recovery time) differ at the levels of a nominal variable with two levels (male v female, producer A v producer B or old drug V new drug)

Suppose that the following data is obtained after running all of the experiments for example 1

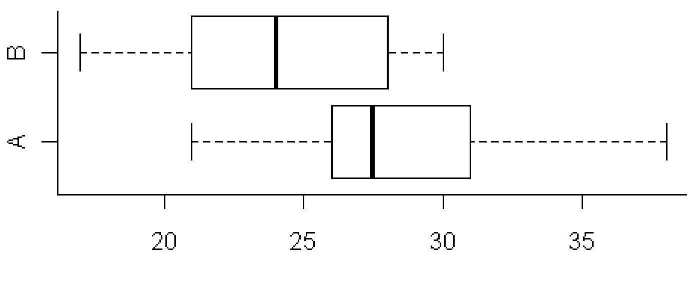
A	26	38	31	29	26	21
B	25	21	30	28	17	23

We might next calculate the following descriptive statistics and construct a clustered boxplot.

```
> summary(A);summary(B)
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   21.0    26.0    27.5    28.5    30.5    38.0
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  17.00   21.50   24.00   24.00   27.25   30.00

> sd(A);sd(B)
[1] 5.75326
[1] 4.732864

> boxplot(A, B, horizontal = TRUE)
```



That the plot for firm A is clearly right shifted relative to B certainly indicates that theirs contain more fat.

Note that 25% of the sample from B has fat content less than the burger from firm B with minimum fat. Also, the median of A is almost the same as the third quartile for B.

Thus, there is clearly a difference in mean fat contents – in the samples.

But does it necessarily follow that there will be differences in the population?

Clearly there is variation in fat content from burger to burger.

Consequently, the sample means for the two producers will if the experiment were repeated (sampling error)

A repeat of the experiment might see a higher sample mean for B than for A.

Hypothesis tests

We present the problem using a hypothesis test,

Let μ_A be the true mean fat content of burgers from firm A – i.e. the mean in the population.

N.B. The true mean is unknown and usually unknowable!

Similarly let μ_B be the true mean fat content of burgers from firm B.

We formulate the null and alternative hypotheses

H_0: $\mu_A = \mu_B$ $\qquad$ H_A: $\mu_A \neq \mu_B$

The appropriate formula (assuming the variation in fat content is the same in the two firms) is

$$\left(\bar{x}_1 - \bar{x}_2\right) \pm t_{n_1+n_2-2}\sqrt{S_P^2\left(\frac{n_1+n_2}{n_1 n_2}\right)} \text{ where } S_P^2 = \frac{(n_1-1)s_1^2+(n_2-1)s_2^2}{n_1+n_2-2}$$

We construct a confidence interval estimate for the difference between μ_A and μ_B.

The following formula may be used to calculate the confidence interval (provided the standard deviations may be assumed to be the same) – but usually, we will let R carry out the calculations.

t two-tailed tables at the 5% significance level

n	1	2	3	4	5	6	7	8	9	10
t	12.71	4.303	3.182	2.776	2.571	2.447	2.365	2.306	2.262	2.228

n	11	12	13	14	15	16	17	18	19	20
t	2.20	2.18	2.16	2.145	2.131	2.120	2.110	2.101	2.093	2.086

We will consider firm A to be sample 1 and B to be sample 2.

Thus $n_1 = 6$ $\;$ $n_2 = 6$ $\;$ $s_1 = 5.75$ $s_2 = 4.73$ $\quad$ $\bar{x}_2 = 24\bar{x}_1 = 28.5$

Note too that $t_{n1+n2-2} = t_{10} = 2.228$ (from the t tables)

Thus, we can evaluate the confidence interval.

$S_p^2 = [9(5.75)^2 + 9(4.73)^2] / 18 = 27.75$ and so $\sqrt{S_P^2\left(\frac{n_1+n_2}{n_1 n_2}\right)} = 3.04$

Thus, our confidence interval is (28.5 – 24) ± 2.228(3.04) = 4.5 ± 6.8 = [4.5 – 6.8, 4.5 + 6.8] = [-2.3, 11.3]

What does this mean?
We don't know the true difference in fat content between firm A and firm B, but we estimate that the difference is somewhere in the interval [-2.3, 11.3]

I.e. We estimate that A might contain as much as 11.3g more fat than B. Or it might contain as much as 2.3g less fat on average.

We can be 95% confident that the difference is in this interval. If we were to correctly construct a large number of such confidence intervals then 95 times out of 100, or 19 times out of 20 on average, our interval estimate would contain the true value. And 1 time in 20 on average, the true value will be outside the calculated interval.

N.B. We cannot know for any given confidence interval construction if it is one of the 95%. Thus, it is possible that the true value is greater than 11.3 or less than -2.3.

So, do fat contents differ?
Our confidence interval-estimate permits that the fat in A might be greater than in B (by as much as 11.3 g) but it also permits that it might be less than in B (by as much as 2.3 g). I.e. the interval contains zero.

Consequently, we do not reject the null hypothesis.

i.e. we are unable to say that there is a difference in mean fat content between the firms.
The higher amount of fat in sample A compared to B could plausibly be explained by sampling error.

A Figure of speech!
You will no doubt have often heard people when asked if they are sure of something or other, replay with something like "I'm 95% certain". This is essentially a figure of speech and could not (should not!) be interpreted as being in anyway meaningful.

An advantage of statistical inference tests <u>properly carried out</u>, over ad-hoc decision making is that while there is still potential for errors to be made, the likelihood of an error can be meaningfully quantified.

The possible errors (there are two types) are conveyed in the table below.

		The (unknown) reality	
		H_0 true	H_0 False
Our conclusion	Do not reject H_0	✓	Type II error
	Reject H_0	Type I error	✓

If H_0 is true and we do not reject H_0 then we have not made an error.
I.e. there is no difference between firm A and B and we did not conclude that there was.

If H_0 is false and we do reject H_0 then again, we have not made an error.
I.e. there is a difference between firm A and B and we did conclude that there was.

If H_0 is true and we reject H_0 then we have made an error – a so called type I error.
I.e. there is no difference between firm A and B but we concluded that there was.

If H_0 is false and we do not reject H_0 then we have made an error – a so called type II error.
I.e. there is a difference between firm A and B but we have not concluded this
As we constructed a 95% confidence interval it follows that the probability of a type I error is 5%

The convention is to fix the probability of a type I error at 5%. The probability of a type II error is often is not considered.

Why 5%

Essentially this is a convention, much as it is a convention that a pass mark in an exam is 40% or the speed limit in an urban area is 50 Kmph.

Using R

We will not usually bother with such tedious formulae as above – we use R to do the calculations.
Assuming the sample data is unstacked. I.e. two separate vectors are used to store A and B.

```
> t.test(A, B, var.equal=TRUE)

        Two Sample t-test

data:  A and B
t = 1.4796, df = 10, p-value = 0.1698
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -2.27662 11.27662
sample estimates:
mean of x mean of y
     28.5      24.0
```

The 95% confidence interval is cited as part of the output.

Also, the mean sample values are calculated for the two groups – we can see that the mean content for x - the first sample (A) is 28.5g and for y – the second sample (B) is 24.0 g.

One other useful statistic included in the output is the p value – here it evaluates to 0.1698. We will see how this is interpreted later.

Note that an argument has been included in the call to `t.test` above to specify that the variances are equal. This is done here, simply to match the output from R with the calculated formula.

Typically, you would not be able to say the variances are equal and thus would use the default, variances differ option as follows.

```
> t.test(A, B)
```

Exercise
Show that in this case, the confidence interval estimate is almost exactly the same.